

SANTIAGO DE COMPOSTELA AP, Spain

WMO: 080420

Lat: 42.888N

Lon: 8.411W

Elev: 370

StdP: 96.96

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.2	1.0	-3.9	2.8	4.0	-2.1	3.3	4.2	11.2	11.2	10.1	10.8	1.5	100	0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.6	30.5	20.1	28.2	19.2	26.1	18.5	21.2	28.2	20.1	26.4	19.2	24.7	2.9	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	14.4	23.6	18.1	13.6	22.1	17.2	12.9	20.7	63.2	28.3	59.3	26.5	56.1	24.2	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	7.8	6.8	-3.0	34.9	1.3	2.1	-4.0	36.4	-4.8	37.6	-5.5	38.8	-6.5	40.4		
(5)			-3.4	23.7	1.3	1.5	-4.3	24.8	-5.0	25.7	-5.7	26.6	-6.7	27.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.1	7.9	8.2	10.3	11.4	13.8	16.9	18.6	19.0	17.3	14.4	10.2	8.6		
	DBStd	4.92	2.60	2.74	3.06	3.28	3.18	3.10	2.82	2.57	2.74	2.89	2.82	2.71		
	HDD10.0	286	74	62	34	21	3	0	0	0	0	2	30	59		
	HDD18.3	2048	323	284	250	208	146	63	30	22	51	125	244	302		
	CDD10.0	1412	9	11	42	63	121	207	268	278	220	139	37	16		
	CDD18.3	133	0	0	0	1	5	21	39	42	21	4	0	0		
	CDH23.3	1271	0	0	4	27	84	234	347	348	189	37	0	0		
	CDH26.7	384	0	0	0	2	18	77	118	111	52	6	0	0		
(14) Wind	WSAvg	3.0	3.1	3.1	3.3	3.2	3.1	2.9	2.9	2.8	2.5	2.7	2.9	3.1		
(15) Precipitation	PrecAvg	1576	210	138	134	132	112	54	39	51	90	218	203	195		
	PrecMax	2105	453	376	582	386	263	137	129	187	262	482	423	594		
	PrecMin	1030	34	0	1	40	30	0	0	1	16	1	11	19		
	PrecStd	318	110	94	127	88	61	42	31	47	64	112	116	127		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.8	19.5	23.9	26.2	29.2	32.8	33.5	33.0	31.2	27.2	19.9	16.2		
		MCWB	11.1	12.0	13.5	15.5	18.0	20.6	20.6	21.4	18.9	17.5	14.7	11.7		
	2%	DB	13.8	16.7	21.1	23.3	25.7	28.9	29.9	29.8	27.8	23.7	17.2	14.4		
		MCWB	11.6	10.8	12.4	14.5	16.8	19.4	20.7	20.1	18.7	16.6	14.0	12.2		
	5%	DB	12.8	14.1	18.4	20.3	22.9	25.9	27.0	27.0	25.2	21.1	15.8	13.6		
		MCWB	11.4	10.0	11.5	13.5	16.0	18.3	19.9	19.4	17.8	15.7	13.7	12.0		
	10%	DB	11.9	12.2	15.6	17.2	19.9	23.2	24.7	24.8	23.0	19.0	14.2	12.8		
		MCWB	10.7	10.2	10.8	12.1	14.7	17.3	18.6	18.6	17.1	15.5	12.7	11.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.6	13.4	14.7	16.7	19.4	22.0	22.5	22.9	20.9	19.3	16.3	14.2		
		MCDB	14.0	16.3	19.1	23.9	26.5	28.8	30.3	30.5	27.9	22.8	17.4	14.5		
	2%	WB	12.7	12.3	13.5	15.1	17.7	20.1	21.2	21.1	19.5	18.0	15.1	13.4		
		MCDB	13.1	14.1	18.0	21.9	24.0	27.2	28.2	27.7	25.6	20.9	16.5	13.8		
	5%	WB	11.8	11.6	12.5	14.0	16.4	18.8	20.0	19.9	18.5	17.0	14.0	12.6		
		MCDB	12.4	13.1	16.6	19.4	21.6	24.9	26.2	25.4	23.6	19.4	15.1	13.2		
	10%	WB	11.0	10.7	11.7	12.8	15.1	17.5	18.9	18.9	17.7	16.1	13.1	11.6		
		MCDB	11.6	12.2	14.9	16.5	18.9	22.4	23.9	23.4	21.8	18.3	14.1	12.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.3	7.7	9.1	9.4	9.6	10.3	10.4	10.6	10.1	8.3	6.7	6.4		
		MCDBR	7.5	11.8	14.8	15.5	15.4	15.8	15.3	15.2	14.6	12.6	8.8	7.5		
		MCWBR	5.6	7.1	7.9	8.1	8.1	7.9	7.4	7.1	6.8	6.6	5.9	5.6		
	5% WB	MCDBR	5.2	8.1	11.3	13.2	13.6	14.0	13.8	13.1	12.5	9.1	6.5	5.3		
		MCWBR	4.6	6.0	6.8	7.3	7.5	7.4	7.1	6.7	6.4	5.7	5.2	4.9		
(40) Clear-Sky Solar Irradiance	taub		0.324	0.330	0.362	0.366	0.371	0.374	0.363	0.365	0.357	0.351	0.330	0.317		
	taud		2.489	2.456	2.372	2.366	2.375	2.383	2.417	2.419	2.445	2.474	2.490	2.501		
	Ebn at Noon		814	869	879	901	902	896	903	891	874	833	797	787		
	Edh at Noon		73	88	108	117	119	118	113	109	99	85	72	66		
(44) All-Sky Solar Radiation	RadAvg		1.50	2.44	3.56	4.80	5.78	6.40	6.43	5.80	4.57	2.86	1.75	1.37		
	RadStd		0.22	0.38	0.56	0.52	0.56	0.44	0.40	0.43	0.30	0.37	0.28	0.20		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+34

Nomenclature: See separate page